Summary in Graph

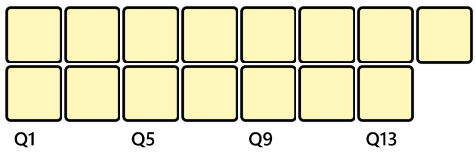
Exam Summary (GO Classes DA Test Series 2026 | Probability and Statistics | Discrete and Continuous Random Variables | Topic Wise Test 3)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**15** (5 + 10)
Total Marks:	**25** (5 + 20)
Exam Duration:	**45 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (76) out of 755 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Given the probability mass function $p(X = x)$ defined as:

$$p(X = x) = \begin{cases} c\left(\frac{2}{3}\right)^x & \text{if } x = 1, 2, 3, \ldots, \infty \\ 0 & \text{otherwise} \end{cases}$$

find the constant c so that $p(X = x)$ is a valid pmf for the random variable X.

A. $c = \frac{1}{3}$
B. $c = \frac{3}{2}$
C. $c = \frac{1}{2}$
D. $c = \frac{2}{3}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The following is the probability distribution for a random variable, X,

k	$\mathrm{P}(X = k)$
0	0.3
1	0.5
2	0.1
3	0.1

What is the variance of X ?

A. 0.800
B. 0.447
C. 0.894
D. 1.225

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Imagine that you do not understand anything in this class. You would have to randomly guess the answer for some given 5 problems (MCQ type with 4 options and only one option is correct). What is the probability that you can get at least two out of five right?

A. $\sum_{k=2}^{5} \left(\frac{1}{4}\right)^k$
B. $\sum_{k=2}^{5} \binom{5}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{5-k}$
C. $\left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^3$
D. $\sum_{k=2}^{5} \frac{5!}{k!} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{5-k}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X_1 and X_2 be two Bernoulli random variables with the probability of success p. These variables are independent, where $X_i = 0$ with probability $1 - p$ and $X_i = 1$ with probability p for $i = 1, 2$. Define a random variable $Y = \frac{X_1 + X_2}{2}$.
What are the correct joint probabilities for $P(Y = 1, X_1 = 1)$ and $P(Y = 0, X_1 = 0)$?

A. $P(Y = 1, X_1 = 1) = p^2, \quad P(Y = 0, X_1 = 0) = 1 - p^2$
B. $P(Y = 1, X_1 = 1) = p^2, \quad P(Y = 0, X_1 = 0) = (1 - p)^2$

C. $P(Y=1, X_1=1) = p(1-p), \quad P(Y=0, X_1=0) = (1-p)^2$
D. $P(Y=1, X_1=1) = 1-p^2, \quad P(Y=0, X_1=0) = p(1-p)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a random variable that takes values from 0 to 9 with equal probability $\frac{1}{10}$. Define the random variable $Y = X \bmod 3$.
Calculate $P(Y=0) + P(Y=1)$.

A. $\frac{3}{10}$
B. $\frac{4}{10}$
C $\frac{5}{10}$
D. $\frac{7}{10}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider the random variables (X, Y) with the joint probability distribution given by:

$$(X,Y) = \begin{cases} (1,4) & \text{with probability } \frac{1}{3} \\ (2,2) & \text{with probability } \frac{1}{3} \\ (4,1) & \text{with probability } \frac{1}{3} \end{cases}$$

Determine the value of the cumulative distribution function $F_{X,Y}$ at the point $(3.5, 4.5)$, i.e., $F_{X,Y}(3.5, 4.5)$.

A. $\frac{1}{3}$
B. $\frac{2}{3}$
C. $\frac{5}{6}$
D. $\frac{1}{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider a discrete random variable X with the following probability mass function (pmf):

$$P(X=x) = \begin{cases} \frac{1}{4} & \text{if } x = 2 \\ \frac{1}{2} & \text{if } x = 4 \\ \frac{1}{4} & \text{if } x = 6 \end{cases}$$

What is the value of $P\left(X^2 - 12X + 35 > 0\right)$?

A. $\frac{1}{4}$
B. $\frac{3}{8}$
C. $\frac{1}{2}$
D. $\frac{3}{4}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Consider the following discrete random variable problem:
A number X is chosen at random from the set $\{1, 2, 3, 4, 5\}$. Subsequently, a second number Y is chosen at random from the subset $\{1, \ldots, X\}$.
Given this setup, which of the following statements is/are true?

A. $\mathbb{P}(X = 5 \mid Y = 5) = \frac{1}{2}$
B. $\mathbb{P}(X = 5, Y = 5) = \frac{1}{25}$
C. $\mathbb{P}(X = 5 \mid Y = 5) = 1$
D. $\mathbb{P}(X = 5, Y = 5) := \frac{1}{20}$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose that n people have their hats returned at random. Let X_i be 1 if the i-th person gets his or her own hat back and 0 otherwise. Find the expected value $E\left(X_i \cdot X_j\right)$ for $i \neq j$.

A. $\frac{1}{n(n-1)}$
B. $\frac{2}{n(n-1)}$
C. $\frac{1}{n^2}$
D. $\frac{1}{n}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 2 | Penalty: 0 | Probability

James Bond is playing a game that involves repeatedly rolling a fair standard 6-sided die. What is the expected number of rolls until he gets a 5 ?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let the joint p.m.f. of X and Y be defined by

$$f(i, j) = \frac{i + j}{32}$$

where $i = 1, 2$ and $j = 1, 2, 3, 4$. Find $P(X \leq 3 - Y)$.

A. $\frac{5}{32}$
B. $\frac{8}{32}$
C. $\frac{9}{32}$
D. $\frac{10}{32}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose you are applying to graduate school and want to improve your GRE writing score. Your score on any given day can be either 4 , 5 , or 6 , each with equal probability. You plan to take the test twice and will report the highest score out of the two tests. Let $X1$ and $X2$ be your scores on the two tests, which are independent of each other, and your final score $X = \max\{X1, X2\}$.
What is the probability that your final score X is 5 , i.e., $P(X = 5)$?

A. $\frac{1}{3}$
B. $\frac{4}{9}$
C. $\frac{5}{9}$
D. $\frac{2}{9}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose X is the number of dust storms that occur on Mars next year. Assume that X is a discrete uniform random variable that takes one of the 101 values in the range $\{0, 1, 2, \ldots, 100\}$. Let $Y = |X - E(X)|$. Find $E(Y)$.

A. $\frac{1275}{101}$
B. $\frac{50}{101}$
C. $\frac{100}{101}$
D. $\frac{2550}{101}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose the discrete random variable X has the probability mass function $f(x) = \frac{11-2x}{24}, \quad x = 1, 2, 3, 4.$
Find $E\left[X^2\right]$.

A. $\frac{110}{24}$
B. $\frac{120}{24}$
C. $\frac{130}{24}$
D. $\frac{140}{24}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 03sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X and Y be independent discrete random variables with probability mass functions $P(X = k)$ and $P(Y = k)$.
What is $P(\min\{X, Y\} \leq x)$?

A. $P(\min\{X, Y\} \leq x) = 1 - P(X > x)P(Y > x)$
B. $P(\min\{X, Y\} \leq x) = P(X \leq x)P(Y \leq x)$
C. $P(\min\{X, Y\} \leq x) = P(X \leq x) + P(Y \leq x) - P(X \leq x)P(Y \leq x)$
D. $P(\min\{X, Y\} \leq x) = P(X > x) + P(Y > x) - P(X > x)P(Y > x)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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